F18CF - Assignment 2

Instructions:

- This assignment is due at 23:59 (Edinburgh Time) on the date specified above
- Your solutions can be written or typeset. No paper submission is necessary
- *Gradescope* submission link available on assignment page in Canvas
- The assessment criteria are listed on the assignment page
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 4 questions worth a total of 40 marks. Marks are listed with each question. An additional 2 marks are awarded for meeting the submission standards.

Questions:

8 marks 1. (a) Consider $\mathcal{P}_2(\mathbb{R})$ with its basis $\mathcal{B} = (x, x+1, x^2-1)$. Find $[x^2+3x+5]_{\mathcal{B}}$.

Solution: Express x^2+3x+5 as a linear combination of polynomials in $\mathcal{B}$:

$$x^2+3x+5 = \alpha_1(x) + \alpha_2(x+1) + \alpha_3(x^2-1)$$
$$x^2+3x+5 = \alpha_3(x^2) + (\alpha_2+\alpha_1)(x) + (\alpha_2-\alpha_3)(1)$$

By comparing coefficient, we have:

$$\begin{aligned} \alpha_3 &= 1 \\ \alpha_2 + \alpha_1 &= 3 \\ \alpha_2 - \alpha_3 &= 5 \end{aligned}$$

Solving the system gives $\alpha_3 = 1, \alpha_2 = 6$ and $\alpha_1 = -3$. As a result, $[x^2+3x+5]_{\mathcal{B}} = (-3, 6, 1)^T$.

(b) Show that $S = \{(a, 0, b)^T | a, b \in \mathbb{R}\}$ forms a subspace of $\mathbb{R}^3$.

Solution: We prove that S fulfills the three conditions of being a subspace.

- **Containing zero-vector**
  Set $a = b = 0$ gives $(0, 0, 0)^T \in S$.
- **Closure under vector addition**
  Let $(a_1, 0, b_1)^T, (a_2, 0, b_2)^T \in S$. We have:

  $$(a_1, 0, b_1)^T + (a_2, 0, b_2)^T = (a_1 + a_2, 0, b_1 + b_2)^T \in S$$

  since $a_1 + a_2, b_1 + b_2 \in \mathbb{R}$.
- **Closure under scalar multiplication**
  Let $(a, 0, b)^T \in S$. We have for every $c \in \mathbb{R}$:

  $$c(a_1, 0, b_1)^T = (c\,a_1, 0, c\,b_1)^T \in S$$

  since $c\,a_1, c\,b_1 \in \mathbb{R}$.

Solution: (Alternate solution) Every vector in S can be written as the sum of two elements in the standard basis: $(a, 0, b)^T = a(1, 0, 0)^T + b(0, 0, 1)^T$ for some $a, b \in \mathbb{R}$. This implies $S = \text{span}((1, 0, 0)^T, (0, 0, 1)^T)$. By Theorem 3.34, we know S is a subspace of $\mathbb{R}^3$.

(c) Determine a basis and the dimension of the following subspace of $\mathbb{R}^4$.

$$V = \text{span}((1, -1, 2, 4)^T, (1, 1, 1, 2)^T, (3, 5, 2, 4)^T, (-3, 3, -6, -12)^T)$$

Solution: Form the following matrix and apply Gaussian Elimination:

$$\begin{pmatrix} 1 & 1 & 3 & -3 \\ -1 & 1 & 5 & 3 \\ 2 & 1 & 2 & -6 \\ 4 & 2 & 4 & -12 \end{pmatrix} \sim \begin{pmatrix} 1 & 1 & 3 & -3 \\ 0 & 2 & 8 & 0 \\ 0 & -1 & -4 & 0 \\ 0 & -2 & -8 & 0 \end{pmatrix} \quad \begin{matrix} R_2 \to R_2 + R_1 \\ R_3 \to R_3 - 2R_1 \\ R_4 \to R_4 - 4R_1 \end{matrix}$$

$$\sim \begin{pmatrix} 1 & 1 & 3 & -3 \\ 0 & 2 & 8 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} \quad \begin{matrix} R_3 \to R_3 + \frac{1}{2}R_2 \\ R_4 \to R_4 + R_2 \end{matrix}$$

The matrix is in Row Echelon Form and has 2 pivot variables correspond to Row 1 and Row 2. Hence, V has dimension 2 and a basis of V is $(1, -1, 2, 4)^T, (1, 1, 1, 2)^T$.

11 marks 2. Let $L : \mathbb{R}^3 \to \mathbb{R}^3$ be a linear map with associated matrix

$$[L]_{\mathcal{B}} = \begin{pmatrix} -1 & 0 & 0 \\ -2 & 2 & -1 \\ 4 & 0 & 1 \end{pmatrix}$$

under the standard basis $\mathcal{B} = ((1,0,0)^T, (0,1,0)^T, (0,0,1)^T)$.

(a) Evaluate $L((2,-1,1)^T)$ under the standard basis

Solution: Using the associated matrix gives:

$$L((2,-1,1)^T) = \begin{pmatrix} -1 & 0 & 0 \\ -2 & 2 & -1 \\ 4 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix} = \begin{pmatrix} -2 \\ -7 \\ 9 \end{pmatrix}$$

(b) Define $\mathcal{C} = \left((0,1,1)^T, (0,1,0)^T, (-1,0,2)^T\right)$. Show that $\mathcal{C}$ forms a basis of $\mathbb{R}^3$.

Solution: Setting up the linear independence equation with the vectors give

$$\alpha_1 \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} + \alpha_2 \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} + \alpha_3 \begin{pmatrix} -1 \\ 0 \\ 2 \end{pmatrix} = \vec{0}$$

which we can convert into an augmented matrix and row reduce:

$$\left(\begin{array}{ccc|c} 0 & 0 & -1 & 0 \\ 1 & 1 & 0 & 0 \\ 1 & 0 & 2 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 1 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right) \qquad R_1 \leftrightarrow R_3$$

$$\sim \left(\begin{array}{ccc|c} 1 & 0 & 2 & 0 \\ 0 & 1 & -2 & 0 \\ 0 & 0 & -1 & 0 \end{array}\right) \qquad R_2 \to R_2 - R_1$$

We have a full set of pivots, which we can back substitute to give $\alpha_1 = \alpha_2 = \alpha_3 = 0$ as the only solution, which shows the vectors are linearly independent. Combined with Theorem 2.24, we have that $\mathcal{C}$ is a basis

Solution: (Alternate solution) We can create a matrix from the vectors to give.

$$\begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

We can compute the determinant by expanding along the first row

$$\det \begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} = -1 \cdot \det \begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}$$

$$= -1 \cdot (1 \times 0 - 1 \times 1) = 1$$

Since the determinant is non-zero Theorem 2.29 tells use that the vectors form a basis.

(c) Find the change of basis matrix $[\mathcal{C}]_{\mathcal{B}}$ and hence find $[\mathcal{B}]_{\mathcal{C}}$.

Solution: From above, we have:

$$[\mathcal{C}]_{\mathcal{B}} = \begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

To find $[\mathcal{B}]_{\mathcal{C}}$, we use the matrix inverse

$$\left(\begin{array}{ccc|ccc} 0 & 0 & -1 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 2 & 0 & 0 & 1 \end{array}\right) \sim \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & -1 & 1 & 0 & 0 \end{array}\right) \qquad R_1 \leftrightarrow R_3$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & 1 & -1 \\ 0 & 0 & -1 & 1 & 0 & 0 \end{array}\right) \qquad R_2 \to R_2 - R_1$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 2 & 0 & 0 & 1 \\ 0 & 1 & -2 & 0 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 0 \end{array}\right) \qquad R_3 \to -1 \cdot R_3$$

$$\sim \left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 2 & 0 & 1 \\ 0 & 1 & 0 & -2 & 1 & -1 \\ 0 & 0 & 1 & -1 & 0 & 0 \end{array}\right) \qquad \begin{array}{r} R_1 \to R_1 - 2R_3 \\ R_2 \to R_2 + 2R_3 \end{array}$$

The right half of the matrix gives us the desired result. $[\mathcal{B}]_{\mathcal{C}} = \begin{pmatrix} 2 & 0 & 1 \\ -2 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix}$

(d) Find $[L]_{\mathcal{C}}$ and hence generalise your result to $([L]_{\mathcal{C}})^N$ for a positive integer N

Solution: Use Theorem 4.39 to get that

$$\begin{aligned} [L]_{\mathcal{C}} &= [\mathcal{B}]_{\mathcal{C}} \times [L]_{\mathcal{B}} \times [\mathcal{C}]_{\mathcal{B}} \\ &= \begin{pmatrix} 2 & 0 & 1 \\ -2 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix} \begin{pmatrix} -1 & 0 & 0 \\ -2 & 2 & -1 \\ 4 & 0 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix} \end{aligned}$$

Since $[L]_{\mathcal{C}}$ is in a nice diagonal form, we can then use matrix multiplication to get that:

$$([L]_{\mathcal{C}})^N = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & -1 \end{pmatrix}^N = \begin{pmatrix} 1^N & 0 & 0 \\ 0 & 2^N & 0 \\ 0 & 0 & (-1)^N \end{pmatrix}$$

(e) Let N a positive integer. Use the result of the previous part to find $([L]_{\mathcal{B}})^N$.

Solution: Since $[L]_{\mathcal{B}} = [\mathcal{C}]_{\mathcal{B}} \times [L]_{\mathcal{C}} \times [\mathcal{B}]_{\mathcal{C}}$ (Theorem 4.39), we have:

$$\begin{aligned}([L]_{\mathcal{B}})^N &= ([\mathcal{C}]_{\mathcal{B}} \times [L]_{\mathcal{C}} \times [\mathcal{B}]_{\mathcal{C}})^N \\ &= ([\mathcal{C}]_{\mathcal{B}} \times [L]_{\mathcal{C}} \times [\mathcal{B}]_{\mathcal{C}}) \times ([\mathcal{C}]_{\mathcal{B}} \times [L]_{\mathcal{C}} \times [\mathcal{B}]_{\mathcal{C}}) \times \ldots \times ([\mathcal{C}]_{\mathcal{B}} \times [L]_{\mathcal{C}} \times [\mathcal{B}]_{\mathcal{C}})\end{aligned}$$

Since $[\mathcal{C}]_{\mathcal{B}} = [\mathcal{B}]_{\mathcal{C}}^{-1}$, some the intermediate terms cancel and we have

$$\begin{aligned}([L]_{\mathcal{B}})^N &= [\mathcal{C}]_{\mathcal{B}} \times ([L]_{\mathcal{C}})^N \times [\mathcal{B}]_{\mathcal{C}} \\ &= \begin{pmatrix} 0 & 0 & -1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2^N & 0 \\ 0 & 0 & (-1)^N \end{pmatrix} \begin{pmatrix} 2 & 0 & 1 \\ -2 & 1 & -1 \\ -1 & 0 & 0 \end{pmatrix} \\ &= \begin{pmatrix} (-1)^N & 0 & 0 \\ 2 - 2 \cdot 2^N & 2^N & 1 - 2^N \\ 2 - 2 \cdot (-1)^N & 0 & 1 \end{pmatrix}\end{aligned}$$

9 marks

3. Let $\vec{e_1} = (1,0,0)^T, \vec{e_2} = (0,1,0)^T, \vec{e_3} = (0,0,1)^T$ be the standard basis of $\mathbb{R}^3$. Consider a linear map $T : \mathbb{R}^3 \to \mathbb{R}^3$ satisfying $T\left((x,y,z)^T\right) = (0,0,0)^T$ whenever $2x - y + z = 0$.

(a) Show that $T\left((\frac{1}{2},0,-1)^T\right) = (0,0,0)^T$ and $T\left((0,1,1)^T\right) = (0,0,0)^T$.

Solution: Note that both $(\frac{1}{2},0,-1)$ and $(0,1,1)$ satisfy $2x-y+z=0$ as $2(\frac{1}{2})-0+1=0$ and $2(0)-1+1=0$.

(b) Justify whether T is an isomorphism using result from (a).

Solution: The linear map T is not injective since $T\left((\frac{1}{2},0,-1)^T\right) = T\left((0,1,1)^T\right)$. Hence, T is not an isomorphism.

(c) Given $T(\vec{e_3}) = (1,2,3)^T$. Find $T(\vec{e_1})$ and $T(\vec{e_2})$ by using result from (a).

Solution: Express $(1,0,0)^T$ as a linear combination of $(0,0,1)^T$ and $(\frac{1}{2},0,-1)^T$. Solving this results in $(1,0,0)^T = 2(0,0,1)^T + 2(\frac{1}{2},0,-1)^T$. Using linearity of T:

$$T\left((1,0,0)^T\right) = 2T\left((0,0,1)^T\right) + 2T\left((\frac{1}{2},0,-1)^T\right) = 2(1,2,3)^T + 2(0,0,0)^T = (2,4,6)^T$$

Similarly, express $(0,1,0)^T$ as a linear combination of $(0,0,1)^T$ and $(0,1,1)^T$. Solving this results in $(0,1,0)^T = -1(0,0,1)^T + 1(0,1,1)^T$. Using linearity of T:

$$T\left((0,1,0)^T\right) = -T\left((0,0,1)^T\right) + T\left((0,1,1)^T\right) = -(1,2,3)^T + (0,0,0)^T = (-1,-2,-3)^T$$

(d) Use your results of $T(\vec{e_1}), T(\vec{e_2}), T(\vec{e_3})$ to determine the image of an arbitrary vector $(x,y,z)^T \in \mathbb{R}^3$ under T.

Solution: For every $(x,y,z)^T \in \mathbb{R}^3$, we can write:

$$\begin{aligned} T\left((x,y,z)^T\right) &= xT\left((1,0,0)^T\right) + yT\left((0,1,0)^T\right) + zT\left((0,0,1)^T\right) \\ &= x(2,4,6)^T + y(-1,-2,-3)^T + z(1,2,3)^T \\ &= (2x - y + z, 2(2x - y + z), 3(2x - y + z))^T \end{aligned}$$

12 marks 4. Consider the set $\mathbb{C}^2 = \left\{ \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} \middle| z_1, z_2 \in \mathbb{C} \right\}$. Define the vectors of $\vec{u}, \vec{v}, \vec{w} \in \mathbb{C}^2$

$$\vec{u} = \begin{pmatrix} 2 \\ 1-i \end{pmatrix} \qquad \vec{v} = \begin{pmatrix} 2i \\ -1+3i \end{pmatrix} \qquad \vec{w} = \begin{pmatrix} -1+i \\ -2+i \end{pmatrix}$$

(a) Evaluate the following two linear combinations of $\vec{u}, \vec{v}, \vec{w}$:

$$\vec{p} = 2\vec{u} - 2\vec{v} + 4\vec{w} \qquad \vec{q} = (2-i)\vec{u} + 2\vec{v} + (-1+i)\vec{w}$$

Solution: We evaluate the linear combinations directly.

$$\begin{aligned}
\vec{p} &= 2\vec{u} - 2\vec{v} + 4\vec{w} \\
&= 2\begin{pmatrix} 2 \\ 1-i \end{pmatrix} - 2\begin{pmatrix} 2i \\ -1+3i \end{pmatrix} + 4\begin{pmatrix} -1+i \\ -2+i \end{pmatrix} \\
&= \begin{pmatrix} 4 \\ 2-2i \end{pmatrix} - \begin{pmatrix} 4i \\ -2+6i \end{pmatrix} + \begin{pmatrix} -4+4i \\ -8+4i \end{pmatrix} \\
&= \begin{pmatrix} (4) - (4i) + (-4+4i) \\ (2-2i) - (-2+6i) + (-8+4i) \end{pmatrix} \\
&= \begin{pmatrix} 0 \\ -4-4i \end{pmatrix}
\end{aligned}$$

$$\begin{aligned}
\vec{q} &= (2-i)\vec{u} + 2\vec{v} + (-1+i)\vec{w} \\
&= (2-i)\begin{pmatrix} 2 \\ 1-i \end{pmatrix} + 2\begin{pmatrix} 2i \\ -1+3i \end{pmatrix} + (-1+i)\begin{pmatrix} -1+i \\ -2+i \end{pmatrix} \\
&= \begin{pmatrix} 4-2i \\ 1-3i \end{pmatrix} + \begin{pmatrix} 4i \\ -2+6i \end{pmatrix} + \begin{pmatrix} -2i \\ 1-3i \end{pmatrix} \\
&= \begin{pmatrix} (4-2i) + (4i) + (-2i) \\ (1-3i) + (-2+6i) + (1-3i) \end{pmatrix} \\
&= \begin{pmatrix} (4-2i) + (4i) + (-2i) \\ (1-3i) + (-2+6i) + (1-3i) \end{pmatrix} \\
&= \begin{pmatrix} 4 \\ 0 \end{pmatrix}
\end{aligned}$$

(b) We expand our understanding of linear independence by stating explicitly the set of scalars involved in linear combinations. For example, the vectors $\vec{u}$, $\vec{v}$, and $\vec{w}$ are *linearly dependent over* $\mathbb{F}$ if there exist scalars $\alpha, \beta, \gamma \in \mathbb{F}$ not all zero such that:

$$\vec{0} = \alpha\vec{u} + \beta\vec{v} + \gamma\vec{w}$$

and the vectors are *linearly independent over* $\mathbb{F}$ if $\alpha = \beta = \gamma = 0$ is the only solution. Determine if $\vec{u}$, $\vec{v}$ and $\vec{w}$ are linearly independent over $\mathbb{R}$.

Solution: To show linear independence, We want to find real numbers $\alpha, \beta, \gamma \in \mathbb{R}$ such that

$$\vec{0} = \alpha\vec{u} + \beta\vec{v} + \gamma\vec{w}$$

Comparing the first coordinate, we get that

$$0 = \alpha(2) + \beta(2i) + \gamma(-1+i)$$

which we can further separate into real and imaginary components, giving

$$\begin{aligned} 0 &= 2\alpha - \gamma && \text{Real} \\ 0 &= 2\beta + \gamma && \text{Imaginary} \end{aligned}$$

Repeating for second coordinate, we get that

$$0 = \alpha(1-i) + \beta(-1+3i) + \gamma(-2+i)$$

which we can further separate into real and imaginary components, giving

$$\begin{aligned} 0 &= \alpha - \beta - 2\gamma && \text{Real} \\ 0 &= \alpha - 3\beta - \gamma && \text{Imaginary} \end{aligned}$$

We can then create an augmented matrix with this information to solve for the unknowns.

$$\begin{aligned}
\left(\begin{array}{ccc|c} 2 & 0 & -1 & 0 \\ 0 & 2 & 1 & 0 \\ 1 & -1 & -2 & 0 \\ 1 & -3 & -1 & 0 \end{array}\right) &\sim \left(\begin{array}{ccc|c} 2 & 0 & -1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & -1 & -\frac{3}{2} & 0 \\ 0 & -3 & -\frac{1}{2} & 0 \end{array}\right) && \begin{array}{l} R_3 \to R_3 - \frac{1}{2}R_1 \\ R_4 \to R_4 - \frac{1}{2}R_1 \end{array} \\
&\sim \left(\begin{array}{ccc|c} 2 & 0 & -1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) && \begin{array}{l} R_3 \to R_3 + \frac{1}{2}R_2 \\ R_4 \to R_4 + \frac{3}{2}R_2 \end{array} \\
&\sim \left(\begin{array}{ccc|c} 2 & 0 & -1 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right) && R_4 \to R_4 + R_3
\end{aligned}$$

With three pivots on three variables, we can back substitute to find that $\alpha = \beta = \gamma = 0$. Hence the vectors are linearly independent.

(c) Determine if $\vec{v}$ and $\vec{w}$ are linearly independent over $\mathbb{C}$.

Solution: We apply the same principles as before, except this time we have to treat $\beta, \gamma \in \mathbb{C}$.

$$\begin{aligned}
\vec{0} &= \beta\vec{v} + \gamma\vec{w} \\
&= (\beta_1 + i\beta_2)\begin{pmatrix} 2i \\ -1+3i \end{pmatrix} + (\gamma_1 + i\gamma_2)\begin{pmatrix} -1+i \\ -2+i \end{pmatrix} \\
&= \begin{pmatrix} -2\beta_2 + 2i\beta_1 \\ (-\beta_1 - 3\beta_2) + i(3\beta_1 - \beta_2) \end{pmatrix} + \begin{pmatrix} (-\gamma_1 - \gamma_2) + i(\gamma_1 - \gamma_2) \\ (-2\gamma_1 - \gamma_2) + i(\gamma_1 - 2\gamma_2) \end{pmatrix} \\
&= \begin{pmatrix} (-2\beta_2 - \gamma_1 - \gamma_2) + i(2\beta_1 + \gamma_1 - \gamma_2) \\ (-\beta_1 + 3\beta_2 - 2\gamma_1 - \gamma_2) + i(3\beta_1 - \beta_2 + \gamma_1 - 2\gamma_2) \end{pmatrix}
\end{aligned}$$

Extracting the real and imaginary parts of each entry gives four equations which all have to equal zero.

$$\begin{aligned} 0 &= 2\beta_1 + \gamma_1 - \gamma_2 \\ 0 &= -2\beta_2 - \gamma_1 - \gamma_2 \\ 0 &= -\beta_1 + 3\beta_2 - 2\gamma_1 - \gamma_2 \\ 0 &= 3\beta_1 - \beta_2 + \gamma_1 - 2\gamma_2 \end{aligned}$$

We then create and augmented matrix and solve for the unknowns

$$\left(\begin{array}{cccc|c} 2 & 0 & 1 & -1 & 0 \\ 0 & -2 & -1 & -1 & 0 \\ -1 & 3 & -2 & -1 & 0 \\ 3 & -1 & 1 & -2 & 0 \end{array}\right) \sim \left(\begin{array}{cccc|c} 2 & 0 & 1 & -1 & 0 \\ 0 & -2 & -1 & -1 & 0 \\ 0 & 3 & -\frac{3}{2} & -\frac{3}{2} & 0 \\ 0 & -1 & -\frac{1}{2} & -\frac{1}{2} & 0 \end{array}\right) \quad \begin{aligned} R_3 &\to R_3 + \tfrac{1}{2}R_1 \\ R_4 &\to R_4 - \tfrac{3}{2}R_1 \end{aligned}$$

$$\sim \left(\begin{array}{cccc|c} 2 & 0 & 1 & -1 & 0 \\ 0 & -2 & -1 & -1 & 0 \\ 0 & 0 & -3 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \quad \begin{aligned} R_3 &\to R_3 + \tfrac{3}{2}R_2 \\ R_4 &\to R_4 - \tfrac{1}{2}R_2 \end{aligned}$$

The row of zeros indicates some linear dependence. In particular, if we let γ_2 be free, we can back-substitute to get $\gamma_1 = -\gamma_2$ in the third row, $\beta_2 = 0$ in the second row, and $\beta_1 = \gamma_2$ from the first row. Since we don't have a full set of pivots, the two vectors are linearly dependent.

In particular, if we let $\gamma_2 = 1$, we find that:

$$\begin{aligned} \vec{v} + (-1+i)\vec{w} &= 0 \\ \vec{v} &= (1-i)\vec{w} \end{aligned}$$

Solution: (Alternate solution) Since we only have two vectors, we can rearrange and see if one is a scalar multiple of the other.

$$\begin{aligned} \vec{0} &= \beta\vec{v} + \gamma\vec{w} \\ \beta\vec{v} &= -\gamma\vec{w} \\ \vec{v} &= -\frac{\gamma}{\beta}\vec{w} \end{aligned}$$

Since $-\frac{\gamma}{\beta} \in \mathbb{C}$, we can write that as: $\vec{v} = -\delta\vec{w}$ for some unknow complex number δ. Substituting and expanding gives:

$$\begin{aligned} \begin{pmatrix} 2i \\ -1+3i \end{pmatrix} &= (\delta_1 + i\delta_2)\begin{pmatrix} -1+i \\ -2+i \end{pmatrix} \\ &= \begin{pmatrix} (-\delta_1 - \delta_2) + i(\delta_1 - \delta_2) \\ (-2\delta_1 - \delta_2) + i(\delta_1 - 2\delta_2) \end{pmatrix} \end{aligned}$$

Extracting the real and imaginary parts gives four conditions

$$\begin{aligned} 0 &= -\delta_1 - \delta_2 \\ 2 &= \delta_1 - \delta_2 \\ -1 &= -2\delta_1 - \delta_2 \\ 3 &= \delta_1 - 2\delta_2 \end{aligned}$$

$$\left(\begin{array}{cc|c} -1 & -1 & 0 \\ 1 & -1 & 2 \\ -2 & -1 & -1 \\ 1 & -2 & 3 \end{array}\right) \sim \left(\begin{array}{cc|c} -1 & -1 & 0 \\ 0 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & -3 & 3 \end{array}\right) \qquad \begin{array}{l} R_2 \to R_2 + R_1 \\ R_3 \to R_3 - 2R_1 \\ R_4 \to R_4 + R_1 \end{array}$$

$$\sim \left(\begin{array}{cc|c} -1 & -1 & 0 \\ 0 & -2 & 2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{array}\right) \qquad \begin{array}{l} R_3 \to R_3 + \frac{1}{2}R_2 \\ R_4 \to R_4 - \frac{3}{2}R_2 \end{array}$$

Row 2 shows that $\delta_2 = -1$ and then substitute into Row 1 to give that $\delta_1 = 1$. Hence we have $\delta = 1 - i$ and

$$\vec{v} = (1 - i)\vec{w}$$

(d) We can create different vector spaces on the same set by using a different choice of scalars. In particular, we can define:

- V to be the vector space of $\mathbb{C}^2$ over $\mathbb{R}$, where the scalars are elements of $\mathbb{R}$
- W to be the vector space of $\mathbb{C}^2$ over $\mathbb{C}$, where the scalars are elements of $\mathbb{C}$

Show that $\vec{e_1} = (1, 0)^T$ and $\vec{e_2} = (0, 1)^T$ forms a basis of W and hence determine the dimension of W

Solution: To show that it forms a basis, we need to show linearly independence and span.

- To show span, we show that $\mathbb{C}^2 \subseteq \text{span}(\vec{e_1}, \vec{e_2})$. Starting with any vector $\vec{v} \in \mathbb{C}^2$, we have

$$\begin{aligned} \vec{v} &= \begin{pmatrix} z_1 \\ z_2 \end{pmatrix} && z_1, z_2 \in \mathbb{C} \\ &= z_1 \begin{pmatrix} 1 \\ 0 \end{pmatrix} + z_2 \begin{pmatrix} 0 \\ 1 \end{pmatrix} \\ &= z_1\vec{e_1} + z_2\vec{e_1} \end{aligned}$$

  Since $\vec{v}$ is a linear combination of the vectors with complex coefficients, we have $\vec{v} \in \text{span}(\vec{e_1}, \vec{e_2})$ As required.

- To show linear independence, we apply the linear independence equation

$$\vec{0} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

  Comparing the two coordinate gives the desired result of $\alpha = \beta = 0$ as the only solution. Therefore, the vectors are linearly independent.

The two vectors form a basis and thus the vector space is of dimension 2

(e) Find a basis of V and hence determine the dimension of V.

Solution: For something to be a basis of V, we have to account for the real and imaginary parts of each component separately. So we propose the set

$$S = \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \begin{pmatrix} i \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ i \end{pmatrix} \right\}$$

To show that it forms a basis, we need to show linearly independence and span.

- To show span, we show that $\mathbb{C}^2 \subseteq \text{span}(S)$. Starting with any vector $\vec{v} \in \mathbb{C}^2$, we have

$$\vec{v} = \begin{pmatrix} a+ib \\ c+id \end{pmatrix} \qquad a, b, c, d \in \mathbb{R}$$
$$= a \begin{pmatrix} 1 \\ 0 \end{pmatrix} + b \begin{pmatrix} i \\ 0 \end{pmatrix} + c \begin{pmatrix} 0 \\ 1 \end{pmatrix} + d \begin{pmatrix} 0 \\ i \end{pmatrix}$$

  Since any vector $\vec{v}$ is a linear combination of the vectors in S with real coefficients, we have $\vec{v} \in \text{span}(S)$

- To show linear independence, we apply the linear independence equation

$$\vec{0} = \begin{pmatrix} 0+i0 \\ 0+i0 \end{pmatrix} = \alpha \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} i \\ 0 \end{pmatrix} + \gamma \begin{pmatrix} 0 \\ 1 \end{pmatrix} + \delta \begin{pmatrix} 0 \\ i \end{pmatrix} = \begin{pmatrix} \alpha + i\beta \\ \gamma + i\delta \end{pmatrix}$$

  Comparing the real and imaginary components of both coordinates first coordinate gives the desired result of $\alpha = \beta = \gamma = \delta = 0$ as the only solution. Therefore, the vectors are linearly independent.

The set S form a basis and thus the vector space is of dimension 4.

(f) Based on your previous results, determine the validity of the following statement:

"V is a subspace of W."

Solution: Theorem 3.39 tells us this statement is invalid. If V is a subspace of W then $\dim(V) \leq \dim(W)$, which isn't the case here.

More generally, this statement doesn't really make mathematical sense. Vector spaces are usually defined over a pre-determined set of scalars and subspaces are only discussed in the context of the same set of scalars. In that sense, Vector spaces V and W are incomparable in this context.